

# JOSEPH IGE IMOLEAYO

Product Engineer · Backend Systems · Distributed Infrastructure

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## PROFESSIONAL SUMMARY

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Backend-focused Product Engineer with production experience in distributed systems, payment infrastructure, and async job processing. I've built a credit-based payment gateway routing transactions across multiple providers with eventual consistency safeguards; isolated file-conversion pipelines where a failed job in one format never affects another; and a health companion backend handling thousands of scheduled reminders with timezone-aware delivery. At Talenvo's Global Residency I was the backend engineer for a cross-functional squad that shipped a live product commended by industry judges at Grand Demo Day. I take end-to-end ownership — from the first ERD to the deployed container.

## TECHNICAL SKILLS

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### Languages & Frameworks

- ▶ Python — FastAPI, Django, Django REST Framework
- ▶ TypeScript / Node.js — Express
- ▶ SQL, Bash, HTML / CSS
- ▶ React (frontend collaboration)

### Architecture & Patterns

- ▶ Distributed systems & microservices
- ▶ Transactional outbox, idempotency patterns
- ▶ Event-driven architecture, WebSockets (Socket.io)
- ▶ REST API design, JWT & SSO auth systems

### Tooling & Workflow

- ▶ Git & GitHub, Postman, VS Code
- ▶ Swagger / OpenAPI documentation
- ▶ Jest — unit & integration testing (27 tests)
- ▶ Vercel, Fly.io, Netlify — cloud deployment

### Data & Infrastructure

- ▶ PostgreSQL, SQLite — schema design & migrations
- ▶ Redis (Upstash) — caching & queue backend
- ▶ RabbitMQ, BullMQ, Celery — async job queues
- ▶ Docker, FFmpeg, Nginx, Prisma ORM

### Payment & Fintech

- ▶ Paystack, OPay — gateway design & routing
- ▶ Credit-based billing & wallet architecture
- ▶ Webhook handling & retry logic
- ▶ Atomic writes, double-charge prevention

### Creative & Multimedia

- ▶ Adobe Photoshop, Illustrator, Premiere Pro
- ▶ After Effects, Blender 3D
- ▶ WordPress, Headless CMS

## EXPERIENCE

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### Backend Engineer · Talenvo Global Residency

2024 – Present · Remote, Lagos, Nigeria

TGR is a selective 10-week program embedding engineers inside product squads tasked with shipping a live, cloud-deployed product under real deadlines and production standards.

- Admitted into the final engineering cohort — a selective process — and ranked in the top 1% of 2,000+ engineers based on assessment output quality and technical depth.
- Served as backend engineer for a cross-functional squad building Vitals, a unified health companion app; contributed significantly to backend architecture, co-authored the system ERD, defined service boundaries, and delivered production APIs consumed directly by the frontend team.

- Built a timezone-aware recurring reminder engine on BullMQ and Redis designed to handle thousands of concurrent scheduled jobs, integrated with Firebase Cloud Messaging for reliable push notification delivery across device types.
- Implemented JWT-based authentication, secure relational database schema design, and RESTful APIs fully documented with Swagger/OpenAPI.
- Owned backend performance under real load; integrated third-party services including payment infrastructure that had to work correctly in production.
- Product shipped live and showcased at Grand Demo Day to an international panel — received strong commendation from industry judges; squad placed in the top tier of all competing teams.
- Built a separate WebSocket-enabled Collaborative Knowledge Board API with Socket.io, Prisma ORM, optimistic locking, and 27 Jest tests — scored 92/100 in blind competitive assessment, 2nd of 6 engineers.
- Diagnosed and resolved a production cross-service auth failure: a Pydantic v2 type-coercion mismatch where Django emitted an integer user ID and FastAPI expected a string under JWT validation — identified and fixed under live conditions.

### **Administrator / Instructor** · [Tech Evolution Academy](#)

*Feb 2023 – Jan 2025*

- Delivered structured technology and digital skills training across multiple cohorts; designed curriculum and learning systems that improved student engagement and measurable outcomes.
- Managed administrative operations, student coordination, and internal workflow processes.

### **Backend Engineering Intern** · [HNG Internship](#)

*2023 · Remote*

- Built and shipped functional APIs under tight timelines; gained practical experience in collaborative engineering workflows, code review, and production-readiness standards.

### **Administrator / Customer Support / Social Media Manager** · [Flip to Tech \(FCMB Partnership\)](#)

*Mar 2022 – Nov 2022*

- Managed administrative operations across academic and non-academic functions; improved operational efficiency through structured workflow redesign and process documentation.

### **Graphic & Motion Designer** · [Freelance](#)

*Oct 2020 – Present*

- Designed visual and motion graphics for clients across multiple industries using Adobe Creative Suite and Blender 3D; delivered branding and communications content to brief and on schedule.

## **PROJECTS**

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### **OfficeSuiteX — Payment Gateway**

<https://github.com/lge-joseph>

*Credit-based payment infrastructure with multi-provider routing, transactional outbox, and wallet consistency safeguards.*

- Built a complete payment gateway accepting payments via OPay and Paystack, routing to the correct provider per transaction and keeping wallet balances accurate even when services are temporarily unavailable.
- Used the transactional outbox pattern — payment records and event messages written atomically, preventing inconsistency between the payment ledger and wallet service.
- Every transaction keyed to a unique idempotency ID; retries are safe with no risk of double-charging.

- Wallet updates run on a separate Django consumer via RabbitMQ — a balance update completes even if the gateway is momentarily down.

[FastAPI](#) [Django](#) [RabbitMQ](#) [SQLAlchemy](#) [PostgreSQL](#) [OPay](#) [Paystack](#)

## OfficeSuiteX — File Conversion Platform

<https://github.com/Ige-Joseph>

*Async file processing infrastructure across 8+ isolated pipelines — documents, images, audio, and video — architected for independent scalability per conversion type.*

- Each of 8+ conversion types runs as its own isolated Celery worker; a failure in a video pipeline never blocks a document conversion — pipelines are independently deployable.
- FFmpeg runs inside Docker containers fully separated from the application layer, so media processing failures stay contained.
- Jobs dispatched asynchronously keep the API non-blocking; adding a new format means adding a new worker, not modifying existing pipelines.

[Python](#) [FastAPI](#) [Docker](#) [FFmpeg](#) [PostgreSQL](#) [Celery](#) [Redis](#)

## Vitals — Health Companion Backend

<https://github.com/Ige-Joseph/vitals-backend>

*Production health companion API supporting shared care, medication reminders, mood tracking, and pregnancy health monitoring.*

- Contributed significantly to backend architecture; co-authored the system ERD and service boundary design before a line of code was written.
- Built a timezone-aware recurring reminder engine on BullMQ and Redis — designed to handle thousands of concurrent scheduled reminders reliably across time zones — integrated with Firebase Cloud Messaging for cross-device push notification delivery.
- Implemented shared care model allowing family members or carers to access and contribute to a user's health profile.
- Delivered modular codebase (controllers / services / repositories) with JWT auth, rate limiting, structured logging, and full Swagger/OpenAPI documentation.

[Node.js](#) [TypeScript](#) [Express](#) [PostgreSQL](#) [Redis](#) [BullMQ](#) [Firebase](#) [Prisma](#)

## Knowledge Board API

<https://github.com/Ige-Joseph/knowledge-board-api>

*Real-time collaborative content platform — WebSocket support, optimistic locking, 27-test Jest suite.*

- WebSocket-enabled real-time collaboration via Socket.io with optimistic locking to handle concurrent edits without data corruption.
- 27 Jest integration tests; scored 92/100 in blind competitive assessment — 2nd of 6 engineers in Talenvo Stage 2 evaluation.

[Node.js](#) [TypeScript](#) [Express](#) [Socket.io](#) [Prisma](#) [PostgreSQL](#) [Jest](#)

## EDUCATION

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### Postgraduate Diploma (PGD) — Information Technology

*Expected 2026 – 2027*

### Bachelor of Science (Education) — Science Education

*National Open University of Nigeria · Second Class Upper*

### Nigeria Certificate in Education (NCE) — Agricultural Science

*Federal College of Education (Technical), Akoka*

## CERTIFICATIONS & RECOGNITION

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- Top 1% — Talenvo Global Residency (2,000+ engineers)
- Grand Demo Day — product shipped live; commended by international panel of industry judges
- 92/100 — Talenvo Stage 2 blind assessment (Knowledge Board API); 2nd of 6 engineers
- Teachers Registration Council of Nigeria (TRCN)
- National Youth Service Corps (NYSC) Certificate